Pairwise Sequence Alignment - Practice Handout

RECAP: Class example

Seq 1: A T G C Seq 2: A T T G C

Scoring scheme: Match = +5 Mismatch = -1 Gap = -2

		A	**T**	**T**	**G**	**C**
	0	-2	-4	-6	-8	-10
A	-2	**5**	**3**	1	-1	-3
T	-4	3	10	**8**	6	4
G	-6	1	8	9	**13**	11
C	-8	-1	6	7	11	**18**

Final alignment:

Remember: Record which direction(s) gave the maximum with an arrow (The blue arrows) .

To get the possible optimal alignment do a traceback from the bottom right **(18)** to the top left cell **(0)** (the red arrows)

The cell with 8 (the green cell), we got the value 8 from two different directions (Horizontal **(10+ -2 =8)** and Diagonal **(3+5 = 8)**)

This means that there are two possible alignments.

Note: When the traceback arrow is diagonal (its either a match or a mismatch of letters), if the traceback arrow is vertical or horizontal, it points towards the sequence that will contain the gap as it pairs with the letter from the other sequence.

Possible Alignment 1:

A – T G C
A T T G C

Possible Alignment 2:

A T – G C
A T T G C

Total alignment score: 18

How to fill the Grid Recap(after the first column and first row are filled)

For each cell (i, j) of the matrix, compute three candidate values and keep the maximum (this is the cell's score).

- **Diagonal, D**: comes from the cell above left, plus the match score if the two letters agree, or the mismatch score if they don't.
- **Vertical, V:** comes from the cell directly above, plus the gap penalty.
- **Horizontal, H:** comes from the cell directly to the left, plus the gap penalty.

Row 0 and column 0 are filled with running totals of the gap penalty before you start. Score(i, j) = max(D, V, H). The traceback starts at the bottom-right cell and ends at the top-left cell (0, 0).

For each question below: fill in every cell of the matrix (value only is fine, but a small arrow for the direction you chose will help you with the traceback), then trace back through the matrix to write out the final aligned sequences and total alignment score.

Place the sequence with more letters at the top of the grid.

Question 1 : Global alignment (Needleman-Wunsch)

Seq 1: G A C T Seq 2: G T C T

Scoring scheme: Match = +1 Mismatch = -1 Gap = -3

Matrix:

Final alignment:

Total alignment score:

Question 2 : Global alignment (Needleman-Wunsch)

Seq 1: T A C G Seq 2: T A C C G

Scoring scheme: Match = +2 Mismatch = -1 Gap = -2

Matrix:

Final alignment:

Total alignment score:

**** Submit your answers before the deadline.****